

# FairScope — AI Fairness Case File

Audit ID: 80690b92-17e  
Target Column: high\_risk

## Verdict: BIASED

Risk Score: 62.0/100 | Confidence: 0.85  
Summary: Based on 4 statistical fairness checks, 2 failed their threshold. Top influencing features were: smoker, insurance\_type, race.

### Key Findings:

- Disparate Impact Ratio on 'race' = 1.424 (PASS)
- Demographic Parity Difference on 'race' = 0.156 (FAIL)
- Disparate Impact Ratio on 'insurance\_type' = 1.688 (PASS)
- Demographic Parity Difference on 'insurance\_type' = 0.229 (FAIL)

### Mitigation Recommendations:

- Apply inverse-probability weighting to balance outcomes across groups in 'smoker'.
- Re-evaluate feature selection: remove or transform features that act as proxies for protected attributes.
- Introduce fairness constraints during model training (e.g., Fairlearn's ExponentiatedGradient).

## Top Feature Importances (Evidence)

| Feature               | Importance | Direction |
|-----------------------|------------|-----------|
| smoker                | 0.16574    | increases |
| insurance_type        | 0.10041    | increases |
| race                  | 0.09566    | increases |
| age                   | 0.06807    | decreases |
| exercise_hours_weekly | 0.04614    | increases |
| cholesterol           | 0.02134    | increases |
| blood_pressure        | 0.02018    | decreases |
| bmi                   | 0.01243    | decreases |

## Fairness Metrics (Compliance)

| Metric                        | Feature        | Value | Passes? |
|-------------------------------|----------------|-------|---------|
| Disparate Impact Ratio        | race           | 1.424 | Yes     |
| Demographic Parity Difference | race           | 0.156 | No      |
| Disparate Impact Ratio        | insurance_type | 1.688 | Yes     |

|                               |                |       |    |
|-------------------------------|----------------|-------|----|
| Demographic Parity Difference | insurance_type | 0.229 | No |
|-------------------------------|----------------|-------|----|